PROJECT EXECUTABLE FILES

ComicCraft – AI Comic Story Creator using Gemini Models

Team ID	SWTID-2026-3296
Team Size	4
Team Leader	Kamalesh P R
Team Members	JAYAKUMARAN V M; Kannan P; Naresh C
Submission Date	15 March 2026 (template date)
Maximum Marks	3 Marks

Prepared using the supplied ComicCraft project description and the Project Executable Files submission template.

Step 1: Submission Checklist

The status below is based on the project documentation supplied. Items requiring actual deployment, repository, or test artifacts are marked accordingly and should be updated before final submission.

S.No	Item to Submit	Status / Details
1	Complete source code (all files and folders)	Referenced by project documentation; attach the complete project source folder separately.
2	README / Setup Guide	Run instructions included in this document; a separate README file is not confirmed.
3	requirements.txt / dependency file	Project documentation refers to requirements.txt and lists required packages. Include the actual file.
4	Database schema / seed files	NA – no database is described in the supplied project details.
5	Environment configuration (.env.example)	Gemini and Hugging Face API key variables are specified; provide a .env.example with placeholders only.
6	Deployed application URL	Not provided; application is documented for local deployment.
7	APK / Executable binary	NA – web application; no mobile/desktop binary described.
8	Dockerfile / Containerization config	Not specified in supplied project details.
9	Test files and test results	Local workflow/API testing is described; separate test files/results were not supplied.
10	Demo video / walkthrough	Not provided.

Step 2: File / Folder Structure

The project documentation identifies the following main modules and resources. Confirm exact paths against the submitted source-code archive; the tree below is a functional overview, not a verified archive listing.

```
ComicCraft/
■■■ app/
■   ■■■ main.py                # FastAPI application entry point (documented run command)
■■■ routes.py                  # FastAPI routes and request handling
■■■ gemini_flash.py            # 5-panel comic outline generation
■■■ gemini_pro.py              # Narration and dialogue generation
■■■ image_generator.py         # Stable Diffusion image generation
■■■ layout_builder.py          # Builds panel/image/story layout
■■■ exporters.py               # Exports comic to PDF using FPDF
■■■ templates/
■   ■■■ index.html             # User prompt form
■   ■■■ comic_preview.html     # Generated comic preview
■   ■■■ export_success.html    # Export confirmation
■■■ static/
■   ■■■ panels/                # Generated panel images
■   ■■■ exports/               # Generated PDF exports
■■■ requirements.txt           # Python dependencies
■■■ .env.example               # Placeholder API-key configuration
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not provided – local access documented at http://127.0.0.1:8000
Login Credentials (Demo)	Not applicable / not described; the documented version does not include user account management.
Platform / Hosting Provider	Local deployment using FastAPI and Uvicorn. Cloud hosting provider not specified.
Repository Link	Not provided.
Demo Video Link	Not provided.

Step 4: Run Instructions

Prerequisites: Python and pip; valid Gemini API key and Hugging Face token; internet access for the AI services. The project uses FastAPI, Uvicorn, Jinja2, Google Generative AI, Hugging Face Diffusers/Transformers, PyTorch, FPDF, Pillow, and Accelerate.

1. Extract or open the complete ComicCraft source folder and open a terminal in the project root.

2. Create and activate a virtual environment:

```
python -m venv env
# Windows
env\Scripts\activate
# macOS/Linux
source env/bin/activate
```

3. Install dependencies (ensure requirements.txt is present):

```
pip install -r requirements.txt
```

4. Configure API credentials as environment variables or in a local .env file loaded by the application. Use your own keys; do not submit real secrets.

```
GEMINI_API_KEY=your_gemini_api_key_here
HF_API_KEY=your-huggingface-api-key-here
```

5. Start the FastAPI server from the project root:

```
uvicorn app.main:app --reload
```

6. Open the application and API documentation in a browser:

```
Application: http://127.0.0.1:8000
API docs:   http://127.0.0.1:8000/docs
```

7. On the homepage, enter a story prompt, character name, setting, tone, and art style. Generate the comic, review the panel-by-panel preview, and use the PDF download option.

Documented endpoints: /, /generate, /generate-comic/json, /export-success, and /test-image.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	The documented version supports comic generation without user account management.	No login is required in the documented workflow.

S.No	Known Issue / Limitation	Workaround / Status
2	Generation depends on external Gemini and Hugging Face services, API credentials, and available resources.	Configure valid API keys and ensure connectivity and sufficient compute resources.
3	A cloud deployment URL, repository, demo video, and standalone test artifacts were not included in the supplied material.	Add the actual links/files if available; otherwise mark as not applicable or not yet provided.

Submission note: This filled template is prepared from the two supplied PDFs. Fields and deliverables not present in those files are explicitly marked “Not provided,” “not specified,” or “confirm” rather than invented.